

MODULAR KOSZUL DUALITY: SECTIONS 2–4

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ABSTRACT. Sections 2–4 develop the curvature calculus of the modular bar construction. Section 2 installs the genus- g propagator via the prime form, computes the period defect, and identifies the single scalar $\kappa(\mathcal{A})$ controlling fibrewise curvature $d_{\text{fib}}^2 = \kappa \cdot \omega_g$. Section 3 develops the modular homotopy theory: the convolution L_∞ -algebra $(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}, \mu_r)$, the modular operad with Getzler–Kapranov Feynman transform, the six-component total differential, and the logarithmic Fulton–MacPherson compactification $\overline{\text{FM}}^{\log}$ of Mok 25 as ambient. Section 4 constructs the universal Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_{\circ} \in \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$, proves $[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ from $D_{\mathcal{A}}^2 = 0$, and records the tridegree, depth, and convergence discipline. The five theorems are projections of this single universal object.

1. CURVATURE AND THE GENUS TOWER

At genus 0 the bar complex is flat: $d^2 = 0$ is an algebraic identity, equivalent to the Borchers axiom. On a family of curves of genus $g \geq 1$, this nilpotence fails. The failure is controlled by a single scalar.

1.1. The prime form and the period defect. On a compact Riemann surface Σ_g of genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning: a meromorphic function with a single simple zero must have a simple pole. The *prime form* $E(z_1, z_2)$, a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing to first order along the diagonal, replaces it. The genus- g propagator

$$\eta_{12}^{(g)} = d \log E(z_1, z_2) + \pi \sum_{\alpha, \beta=1}^g (\text{Im } \Omega)_{\alpha\beta}^{-1} \omega_{\alpha}(z_1) \bar{\omega}_{\beta}(z_2) \quad (1.1)$$

is single-valued; the period correction couples to the Hodge bundle $\mathbb{E} = R^0 \pi_* \omega_{\pi}$ through the period matrix Ω .

Three properties of (1.1) constrain the genus tower. First, the logarithmic derivative $d \log E$ has conformal weight 1 in both variables, regardless of the conformal weight of the field being sewed: every edge of every graph sum at every genus uses the standard Hodge bundle $\mathbb{E}_1$. Second, the period correction is harmonic in each variable and involves $(\text{Im } \Omega)^{-1}$, the Arakelov Green function on Σ_g . Third, the Arnold relation acquires a defect proportional to the Arakelov form, and this defect propagates into the bar differential.

1.2. Curvature. Let $\pi: C_g \rightarrow \overline{\mathcal{M}}_g$ be the universal curve. On each fibre Σ_g , the propagator $\eta^{(g)}$ is single-valued but no longer closed: the period correction has a $\bar{\partial}$ -component, and the Arnold relation fails. The fibrewise bar differential satisfies

$$d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g, \quad (1.2)$$

where ω_g is the Arakelov $(1, 1)$ -form on the fibre and $\kappa(\mathcal{A}) \in \mathbb{C}$ is the *modular characteristic*. The modular characteristic is intrinsic to the chiral algebra; it is determined by the leading OPE singularity and is visible already at genus 0 as the Σ_2 -coinvariant scalar shadow of the collision residue: for abelian

and Virasoro-type families $\kappa(\mathcal{A}) = \text{av}(r(z))$, while for non-abelian affine Kac–Moody in the trace-form convention $\text{av}(r(z)) = \kappa_{\text{dp}}(\mathcal{A})$ and $\kappa(\mathcal{A}) = \kappa_{\text{dp}}(\mathcal{A}) + \dim(\mathfrak{g})/2$.

The failure of $d^2 = 0$ at genus $g \geq 1$ is not a defect of the construction; it is the construction detecting global geometry. Period integrals restore nilpotence: the Gauss–Manin connection absorbs the fibrewise curvature, and the total differential satisfies $D_g^2 = 0$. The curvature (1.2) is the infinitesimal avatar of this restoration.

1.3. The genus expansion. For a modular Koszul chiral algebra $\mathcal{A}$ with a single generator (or more generally with all generators of the same conformal weight), the genus- g obstruction class $\text{obs}_g(\mathcal{A})$ is proportional to the top Chern class of the Hodge bundle:

$$\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g, \quad g \geq 1.$$

The free energy $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$, where

$$\lambda_g^{\text{FP}} = \frac{2^{2g-1} - 1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$$

is the Faber–Pandharipande coefficient (B_{2g} the $2g$ -th Bernoulli number), packages the entire genus tower into the Hirzebruch $\hat{A}$ -class:

$$\sum_{g=1}^{\infty} F_g(\mathcal{A}) \lambda_g^{-2} = \frac{\kappa(\mathcal{A})}{2} [\hat{A}(i) - 1], \quad \hat{A}(x) = \frac{x/2}{\sinh(x/2)}. \quad (1.3)$$

At genus 1: $\lambda_1^{\text{FP}} = 1/24$. At genus 2: $\lambda_2^{\text{FP}} = 7/5760$. The universal ratio $F_2/F_1 = 7/240$ is independent of $\mathcal{A}$. The series converges for $|x| < 2\pi$, the first zero of $\sinh(x/2)$.

The genus-1 identity $\text{obs}_1 = \kappa \cdot \lambda_1$ is unconditional: it holds for all modular Koszul chiral algebras, regardless of the number or weights of their generators. At genus $g \geq 2$, the story is more delicate.

Remark 1.1 (Multi-weight correction). For a chiral algebra with generators of distinct conformal weights $h_1, \dots, h_r$, the genus- g free energy decomposes as

$$F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}(\mathcal{A}),$$

where $\delta F_g^{\text{cross}}$ is a graph sum over mixed-channel boundary graphs of $\overline{\mathcal{M}}_{g,0}$. At genus 1, $\delta F_1^{\text{cross}} = 0$ unconditionally. For uniform-weight algebras ($h_1 = \dots = h_r$), $\delta F_g^{\text{cross}} = 0$ at all genera. For multi-weight algebras at $g \geq 2$, the correction is generically nonzero: for $\mathcal{W}_3$ at genus 2, $\delta F_2^{\text{cross}}(\mathcal{W}_3) = (c + 204)/(16c) > 0$ for all $c > 0$.

1.4. The modular characteristic. The modular characteristic $\kappa(\mathcal{A})$ is the single scalar controlling the genus tower. It is computed from the leading OPE singularity and fixed universally by the genus-1 curvature extracted through the bar construction.

The modular characteristic is additive under independent sums ($\kappa(\mathcal{A} \oplus \mathcal{A}') = \kappa(\mathcal{A}) + \kappa(\mathcal{A}')$ when the mixed OPE vanishes) and constrained by Koszul duality: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger) = 0$ for Kac–Moody and free-field algebras; $\kappa(\text{Vir}_c) + \kappa(\text{Vir}_{26-c}) = 13$ for Virasoro. In general, $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger) = \rho \cdot K$ for $\mathcal{W}$ -algebras, where ρ is the anomaly ratio and K the complementarity constant.

At the self-dual point ($c = 13$ for Virasoro), the complementarity sum is symmetric: $\kappa(\text{Vir}_{13}) = 13/2 = \kappa(\text{Vir}_{13}^\dagger)$. At the critical level ($k = -h^\vee$ for Kac–Moody), $\kappa = 0$ and the bar complex is flat at all genera: the content migrates from curvature to bar cohomology, where it appears as the Feigin–Frenkel centre.

2. THE FIVE MAIN THEOREMS

The bar construction of a chiral algebra on a curve produces a factorization coalgebra on the Ran space of the curve. The five main theorems describe what this coalgebra determines: the Koszul dual,

TABLE 1. Modular characteristic for the standard families

Algebra $\mathcal{A}$	$\kappa(\mathcal{A})$	Shadow depth class
Heisenberg $\mathcal{H}_k$	k	G
Affine Kac–Moody $\widehat{\mathfrak{g}}_k$	$\frac{\dim(\mathfrak{g}) \cdot (k + h^\vee)}{2h^\vee}$	L
$\beta\gamma$ system	$c/2$	C
Virasoro Vir_c	$c/2$	M
$\mathcal{W}_3^k(\mathfrak{sl}_3)$	$5c/6$	M
$\mathcal{W}_N^k(\mathfrak{sl}_N)$	$c \cdot \sum_{j=2}^N \frac{1}{j}$	M

the original algebra, the complementarity decomposition, the genus tower, and the chiral Hochschild cohomology. Each theorem is a projection of the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ (Section 3).

2.1. Theorem A: Verdier intertwining.

Theorem 2.1 (A). *Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X . There is an adjunction*

$$B : \text{Alg}_{\text{aug}}(\text{Fact}(X)) \rightleftarrows \text{CoAlg}_{\text{conil}}(\text{Fact}(X)) : \Omega$$

between augmented factorization algebras and conilpotent factorization coalgebras, with unit and counit the universal twisting morphisms. Verdier duality on $\text{Ran}(X)$ intertwines the bar of $\mathcal{A}$ with the bar of its Koszul dual:

$$\mathbb{D}_{\text{Ran}}(B(\mathcal{A})) \simeq B(\mathcal{A}^!), \quad (2.1)$$

as factorization coalgebras, or equivalently $\mathbb{D}_{\text{Ran}}(B(\mathcal{A})) \simeq \mathcal{A}_{\infty}^{\text{l,co}}$ and $\Omega_X^{\text{cont}} \mathcal{A}_{\infty}^{\text{l,co}} \simeq \mathcal{A}_{\infty}^!$ as the homotopy Koszul dual algebra after completed cobar.

The adjunction is formal: it requires only the augmented factorization-algebra structure and the duality on $\text{Ran}(X)$. The intertwining (2.1) is a deeper statement: it says that the Verdier dual of the bar coalgebra is the bar of the Koszul dual algebra. The identification of the homotopy Koszul dual $\mathcal{A}_{\infty}^!$ (retaining full chain-level data) with the strict Koszul dual $\mathcal{A}^! = (H^*(B(\mathcal{A})))^{\vee}$ (retaining only cohomology) is the content of Theorem A on the Koszul locus, not a tautology.

2.2. Theorem B: bar-cobar inversion.

Theorem 2.2 (B). *On the Koszul locus, the cobar of the bar recovers the original algebra:*

$$\Omega(B(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}.$$

The counit of the bar-cobar adjunction is a quasi-isomorphism of augmented factorization algebras. At genus 0 this is unconditional. For arbitrary modular pre-Koszul inputs at $g \geq 1$, the ordinary direct-sum statement holds only on the Koszul/collapse loci. Off that locus, and for class M , inversion is a weight-completed coderived Positselski statement.

Bar-cobar inversion is a round-trip, not a duality. It does not produce the Koszul dual (that is Verdier duality, Theorem A); it does not produce the bulk algebra (that is the chiral derived centre, Theorem H). It recovers $\mathcal{A}$ itself.

The proof proceeds by constructing a contracting homotopy for the counit, using the PBW filtration as a control parameter: the associated graded of the cobar complex coincides with a cofree resolution of the PBW-graded algebra, and the filtration spectral sequence degenerates at E_2 . Convergence of the spectral sequence requires the Koszul hypothesis; for non-Koszul algebras, bar-cobar inversion requires completion.

2.3. Theorem C: complementarity.

Theorem 2.3 (C). *For a Koszul pair $(\mathcal{A}, \mathcal{A}^\dagger)$ on X , Theorem C has three typed tiers. Co is the strict flat fibre-centre comparison giving the ambient complex*

$$\mathbf{C}_g(\mathcal{A}) := R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$$

on the chirally Koszul datum. C1, once the represented Verdier involution, perfectness, and nondegenerate anti-invariant Verdier pairing are supplied, decomposes this complex into homotopy eigenspaces:

$$\mathbf{C}_g \simeq \mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^\dagger).$$

For $g \geq 1$ the pairing has cohomological degree $-(3g-3)$, and the summands $\mathbf{Q}_g(\mathcal{A}) = \text{fib}(\sigma - \text{id})$ and $\mathbf{Q}_g(\mathcal{A}^\dagger) = \text{fib}(\sigma + \text{id})$ are Lagrangian under the C1 hypotheses. C2 is the shifted-symplectic/BV upgrade; it is conditional on the C2 hypothesis package and is not supplied by the scalar conductor or by uniform-weight diagonalization alone.

The scalar identity $F_g = \kappa \lambda_g^{\text{FP}}$ is the uniform-weight scalar trace shadow of C1 together with Theorem D. It is not the C2 shifted-symplectic upgrade. The C2 statement requires the Pantev–Toën–Vaquié–Vezzosi shifted-symplectic package, cyclic closedness, skew-adjointness, modular-bracket compatibility, twisted-tensor acyclicity, and the stated BV/QME anomaly-killing inputs.

2.4. Theorem D: the modular characteristic.

Theorem 2.4 (D). *For a uniform-weight modular Koszul chiral algebra $\mathcal{A}$, the genus- g obstruction class satisfies $\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g(\text{UNIFORM-WEIGHT})$ for all $g \geq 1$, where $\kappa(\mathcal{A})$ is a universal, additive, duality-constrained scalar fixed by the genus-1 curvature and already visible in the leading OPE singularity of $\mathcal{A}$. The generating function is the Hirzebruch $\hat{A}$ -class ((1.3)). At genus 1, the identity is unconditional: it holds for all modular Koszul algebras regardless of weight structure. At $g \geq 2$ for multi-weight algebras, the full free energy acquires the explicit cross-channel correction $\delta F_g^{\text{cross}}$.*

The uniform-weight scalar lane is the point: after scalar diagonal projection, the genus tower is controlled by the single number $\kappa(\mathcal{A})$. Off that lane the full free energy includes $\delta F_g^{\text{cross}}$. Theorem D is the all-genera identity on the uniform-weight lane; at genus 1 it is unconditional (Remark 1.1).

2.5. Theorem H: chiral Hochschild cohomology.

Theorem 2.5 (H). *Under the Theorem H package H_H (PBW chiral Koszulness, finite-type/perfectness, genericity, E_∞ -completion, strict Mittag–Leffler passage, and localized bar-concentration), the chiral Hochschild cohomology of the generic Virasoro, principal $\mathcal{W}$ -algebra, and affine Kac–Moody families is concentrated in degrees $\{0, 1, 2\}$ with degree- ≤ 2 Hilbert polynomial. For affine Kac–Moody at generic level, after the zero-mode inner quotient,*

$$\text{ChirHoch}^1(V_k(\mathfrak{g})) = 0, \quad P_{V_k(\mathfrak{g})}(t) = 1 + t^2.$$

The adjoint $\mathfrak{g}$ is retained as zero-mode derivation prequotient and bar-dual generator count. At critical level, the comparison belongs to the separate critical-level lane and the cohomology is not bounded by this generic concentration statement.

The chiral derived centre $Z_{\text{ch}}^{\text{der}}(\mathcal{A}) = C_{\text{ch}}^{\bullet}(\mathcal{A}, \mathcal{A})$ is the universal closed-sector algebra: the cochain-level operator algebra acting on the boundary algebra $\mathcal{A}$. A physical bulk interpretation requires the open–closed comparison datum. Theorem H constrains its cohomology. For the Virasoro algebra, the total dimension of $\text{ChirHoch}^*(\text{Vir}_c)$ is at most 4; this is a finite-dimensionality statement, not a polynomial-ring statement.

2.6. The Koszulness meta-theorem. The hypothesis shared by Theorems A–H is chiral Koszulness: bar cohomology $H^*(B(\mathcal{A}))$ concentrated in bar degree 1. The meta-theorem organises this condition by twelve tests and consequences: seven independent bidirectional equivalences, one listed consequence, two conditional comparisons, one one-way chiral Hochschild consequence, and one one-directional purity implication.

Theorem 2.6 (Koszulness characterization). *For a chiral algebra $\mathcal{A}$ on X with PBW filtration, conditions (i)–(v) and (ix)–(x) are the seven independent bidirectional equivalences. Item (vi) follows from (v). Item (vii) is conditional on the factorization-homology comparison and detection package. Item (viii) is a one-way consequence under the Theorem H package H_H : PBW chiral Koszulness, finite-type/perfectness, genericity, E_{∞} -completion, strict Mittag–Leffler passage, and localized bar-concentration. Under the shifted-symplectic hypothesis package, (xi) is equivalent. Condition (xii) implies (x); the converse is open.*

- (i) $\mathcal{A}$ is chirally Koszul.
- (ii) The PBW spectral sequence collapses at E_2 .
- (iii) The minimal A_{∞} -model of $B(\mathcal{A})$ is formal: $m_n = 0$ for $n \geq 3$.
- (iv) Ext diagonal vanishing: $\text{Ext}_{\mathcal{A}}^{p,q}(\omega_X, \omega_X) = 0$ for $p \neq q$.
- (v) The bar-cobar counit $\Omega(B(\mathcal{A})) \rightarrow \mathcal{A}$ is a quasi-isomorphism.
- (vi) The Barr–Beck–Lurie comparison is an equivalence.
- (vii) Factorization homology $\int_{\Sigma_g} \mathcal{A}$ is concentrated in degree 0 for all g .
- (viii) Under H_H , $\text{ChirHoch}^*(\mathcal{A})$ has amplitude $[0, 2]$ and Hilbert polynomial in t .
- (ix) The Kac–Shapovalov determinant is nonzero in the bar-relevant range.
- (x) Restriction to every FM boundary stratum is acyclic outside degree 0.
- (xi) Lagrangian criterion (conditional).
- (xii) $\mathcal{D}$ -module purity (one-directional: $(x) \Rightarrow (xii)$).

The core circuit of independent equivalences runs through PBW degeneration (commutative algebra), A_{∞} -formality (homotopical algebra), Ext diagonal vanishing, bar-cobar inversion, Kac–Shapovalov nonvanishing (representation theory), and FM boundary acyclicity (algebraic topology). The Barr–Beck–Lurie row records the higher-categorical consequence of bar-cobar inversion; the factorization-homology row needs its comparison and detection hypotheses.

3. THE UNIVERSAL MAURER–CARTAN ELEMENT AND THE SHADOW TOWER

The five theorems of Section 2 are projections of a single algebraic object: the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ of the modular convolution algebra.

3.1. The bar-intrinsic construction. The genus-completed bar differential $D_{\mathcal{A}} = \sum_{g \geq 0} g d_{\mathcal{A}}^{(g)}$ satisfies $D_{\mathcal{A}}^2 = 0$ by the codimension-2 face cancellation on $\overline{\mathcal{M}}_{g,n}$ (the same mechanism as §??, extended to all genera by Mok’s log Fulton–MacPherson compactification). Write $d_0 = d_{\mathcal{A}}^{(0)}$ for the genus-0 bar differential.

Construction 3.1 (Bar-intrinsic MC element). Define the positive-genus correction

$$\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o = \sum_{g \geq 1} g d_{\mathcal{A}}^{(g)} \in \prod_{g \geq 1} \text{CoDer}^{\text{cyc}}(B_X^{(g)}(\mathcal{A}))[1].$$

Theorem 3.2. $\Theta_{\mathcal{A}}$ is a Maurer–Cartan element of the modular cyclic deformation complex:

$$d_o(\Theta_{\mathcal{A}}) + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0. \quad (3.1)$$

Proof sketch. Expand $D_{\mathcal{A}}^2 = (d_o + \Theta_{\mathcal{A}})^2 = 0$. Since $d_o^2 = 0$ (genus-0 bar flatness), the remaining terms give $d_o(\Theta_{\mathcal{A}}) + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$, which is the MC equation. The bracket is the convolution bracket on the modular cyclic deformation complex, inherited from stable-curve gluing; its square-zero property is a formal consequence of $d_o^2 = 0$. $\square$

The construction requires no restriction to simple Lie symmetry, no one-channel hypothesis, and no tautological-line support condition. It works for all modular Koszul chiral algebras. The modular characteristic κ , the cubic shadow $\mathfrak{C}$, and the quartic resonance class $\mathfrak{Q}$ are degree projections of this single element:

$$\kappa(\mathcal{A}) = \pi_2(\Theta_{\mathcal{A}}), \quad \mathfrak{C}(\mathcal{A}) = \pi_3(\Theta_{\mathcal{A}}), \quad \mathfrak{Q}(\mathcal{A}) = \pi_4(\Theta_{\mathcal{A}}).$$

Each degree projection carries more information than the last; the full $\Theta_{\mathcal{A}}$ carries all of it.

3.2. The shadow obstruction tower. The degree- r projection $\text{Sh}_r(\mathcal{A}) := \pi_r(\Theta_{\mathcal{A}})$ is the r -th shadow coefficient. The sequence $S_2 = \kappa, S_3 = \alpha, S_4, S_5, \dots$ constitutes the *shadow obstruction tower*. On any one-dimensional primary slice L of the cyclic deformation complex, the MC equation (3.1) imposes a quadratic constraint.

Theorem 3.3 (Riccati algebraicity). *The weighted shadow generating function $H(t) := \sum_{r \geq 2} r S_r t^r$ satisfies*

$$H(t)^2 = t^4 Q_L(t),$$

where

$$Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2, \quad \Delta := 8\kappa S_4,$$

is the shadow metric: a quadratic polynomial in t determined by three genus-0 invariants (κ, α, S_4) . The entire shadow obstruction tower on L is determined by Q_L .

The generating function $H(t) = t^2 \sqrt{Q_L(t)}$ is algebraic of degree 2. The shadow obstruction tower is neither a Taylor expansion that might diverge nor a formal power series requiring analytic continuation: it is the expansion of an explicit algebraic function. The MC equation, which a priori is an infinite system of coupled nonlinear equations, reduces on each primary slice to a single quadratic identity.

3.3. The G/L/C/M classification. The discriminant $\Delta = 8\kappa S_4$ of the shadow metric forces a universal dichotomy.

Definition 3.4 (Shadow depth classification). The *shadow depth* of $\mathcal{A}$ is $r_{\max}(\mathcal{A}) := \sup\{r \geq 2 : S_r \neq 0\}$. The shadow depth class is:

Class	Name	$r_{\max}$	Mechanism
G	Gaussian	2	Q_L perfect square, $\alpha = 0$
L	Lie	3	Q_L perfect square, $\alpha \neq 0$
C	Contact	4	stratum separation
M	Mixed	∞	$\Delta \neq 0$: Q_L irreducible

The mechanism: when $\Delta = 0$, the shadow metric is a perfect square $Q_L(t) = (2\kappa + 3\alpha t)^2$ and $\sqrt{Q_L}$ is polynomial: the tower terminates. If $\alpha = 0$ it terminates at degree 2 (class **G**: Heisenberg); if $\alpha \neq 0$ at degree 3 (class **L**: affine Kac–Moody). When $\Delta \neq 0$, $\sqrt{Q_L}$ is irrational: the tower runs forever (class **M**: Virasoro, $\mathcal{W}_N$). Class **C** ($\beta\gamma$) escapes the dichotomy by stratum separation: the quartic contact invariant lives on a charged stratum whose self-bracket exits the complex by rank-one rigidity.

The four classes are a homotopy-invariant classification: they coincide with the L_∞ -formality depth of the transferred minimal model. Class **G** is L_∞ -formal (all higher brackets vanish). Class **L** has exactly one nontrivial Massey product. Class **C** has formality obstructions through degree 4. Class **M** is intrinsically non-formal: no finite truncation of the L_∞ structure is quasi-isomorphic to a dg Lie algebra.

Archetypes:

Class	Archetype	κ	Δ
G	Heisenberg $\mathcal{H}_k$	k	0
L	$\widehat{\mathfrak{g}}_k$	$\dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$	0
C	$\beta\gamma$	$c/2$	0 (stratum separation)
M	Vir_c	$c/2$	$40/(5c+22)$

For the Virasoro algebra, the quartic contact invariant is $Q_{\text{Vir}}^{\text{contact}} = 10/[c(5c+22)]$ and the critical discriminant is $\Delta_{\text{Vir}} = 8 \cdot (c/2) \cdot 10/[c(5c+22)] = 40/(5c+22)$. Every shadow coefficient S_r for $r \geq 5$ is a rational function of c with poles only at $c = 0$ and $c = -22/5$:

$$S_5 = -\frac{48}{c^2(5c+22)}, \quad S_6 = \frac{80(45c+193)}{3c^3(5c+22)^2}, \quad S_7 = -\frac{2880(15c+61)}{7c^4(5c+22)^2}.$$

These are the fingerprints of a class-**M** algebra: the shadow tower is infinite, alternating in sign, and controlled by an irrational algebraic function.

3.4. The gauge-gravity dichotomy. The classification has a physical interpretation. At $\kappa = 0$ the bar complex is flat at all genera: the genus tower carries no curvature, the free energy vanishes, and the content lives in bar cohomology. This is the regime of gauge theory: the Feigin–Frenkel centre (the Langlands dual of the affine algebra at critical level) is the genus-0 bar cohomology $H^0(B(\widehat{\mathfrak{g}}_{-h^\vee}))$.

At $\kappa \neq 0$ the bar complex is curved: every genus contributes a nonzero obstruction class, the genus tower is nontrivial, and the free energy is the $\hat{A}$ -genus. This is the regime of gravity: the bar curvature is the perturbative genus expansion.

The Koszul duality $\mathcal{A} \leftrightarrow \mathcal{A}^!$ exchanges gauge and gravity sectors. For affine Kac–Moody at generic level, $\kappa(\widehat{\mathfrak{g}}_k) \neq 0$ and the algebra is curved; at the Koszul-dual critical level $k' = -k - 2h^\vee$, $\kappa(\widehat{\mathfrak{g}}_{k'}) \neq 0$ as well, and both sides are gravitational. The gauge regime (critical level, $\kappa = 0$) is the fixed locus of a one-parameter family.

For the Virasoro algebra, the Koszul dual is $\text{Vir}_c^! = \text{Vir}_{26-c}$. The gauge locus is $c = 0$ (topological: $\kappa = 0$); the gravitational locus is $c \neq 0$. The self-dual point $c = 13$ ($\kappa = 13/2 = \kappa^!$) is the critical string. The Clifford algebra of the complementarity pairing degenerates at $c = 26$ ($\kappa(\text{Vir}_0^!) = 0$): the 26-dimensional bosonic string is the critical point where the Koszul dual becomes flat.

The dichotomy gauge/ $\kappa=0$ vs gravity/ $\kappa \neq 0$ is not a physical input: it is a theorem about the bar complex. The physical interpretation follows from the identification of bar curvature with perturbative genus expansions in quantum field theory.